RAVI MOHAN ROHELA

SOFTWARE ENGINEER & FULL-STACK DEVELOPER

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PROFESSIONAL SUMMARY

Software Engineer and Full-Stack Developer with strong expertise in building reliable backend systems, scalable RESTful APIs, and responsive, performance-driven web applications. Demonstrated capability in architecting full-cycle production applications with modern JavaScript/TypeScript ecosystems, Python/Django, relational and document databases, and interactive user experiences.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL, PHP, HTML5, CSS3
Frontend: React, Next.js (App Router), Tailwind CSS, Canvas API, Responsive Design
Backend & APIs: Node.js, Express.js, Django, REST APIs, WebSockets, WebRTC, JWT Auth
Databases: MongoDB, MongoDB Atlas, PostgreSQL, MySQL, Redis
Tools & Platforms: Git, GitHub, Docker, Postman, Linux, Vercel, Render, Vite, VS Code

FEATURED PROJECTS

Quantara - Online Examination & Proctoring Platform *Next.js, Node.js, Express, MongoDB Atlas, WebRTC, Tailwind CSS*

- Built an automated assessment engine with timed questionnaires, role-based dashboards, and instant scoring telemetry.
- Engineered real-time browser anti-cheat mechanisms featuring tab-switch event listeners and video proctoring protocols.
- Designed low-latency REST endpoints handling candidate state persistence and encrypted result exports.

PassOP - Encrypted Password Management Application *React, Node.js, Express, MongoDB Atlas, CryptoJS, Tailwind CSS*

- Developed a zero-knowledge credential vault implementing client-side AES-256 encryption before transmission.
- Created responsive UI with instant search filtering, generator for cryptographically secure passwords, and clipboard copying.
- Secured database persistence with schema-level validation and hashed master credentials.

Travio - Collaborative Travel Planning Platform *React, Node.js, Express, MongoDB Atlas, Mapbox GL, Tailwind CSS*

- Architected collaborative trip management platform with interactive multi-stop route visualizations and Mapbox integration.
- Implemented group expense-splitting module automating multi-currency calculations and individual settlement balances.
- Deployed scalable backend with rate limiting, JWT token rotation, and geospatial query indexing.

Cinemiq - AI-Powered Generative Media & Video Studio *Next.js, React, Node.js, FFmpeg, Canvas API, Tailwind CSS*

- Engineered automated scene timeline composer enabling prompt-driven media orchestration and transitions.
- Implemented canvas-based multi-track real-time preview player with synchronized audio overlays and title cards.
- Structured cloud rendering queue supporting automated aspect ratio conversions and export optimization.

Tune Tide - Web Audio Player & Playlist Curator *React, Next.js, Web Audio API, Tailwind CSS*

- Built a high-performance audio engine with real-time frequency visualizer via HTML5 Canvas and Web Audio API.
- Crafted custom playlist management with drag-and-drop sequencing and resilient LocalStorage offline caching.

EDUCATION

Bachelor of Technology (B.Tech) - Information Technology 2021 - 2025
Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad, Uttar Pradesh, India